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BATTERY CHARGER ENCLOSURE HAVING RETRACTING MECHANISM

Abstract

Enclosures for battery chargers for individual EVs are provided. An enclosure can be configured to house a battery charger and can be secured to a wall near an outlet or other source of electricity. The enclosure can include a retracting mechanism such as a badge reel that can be secured to the charging cable of the battery charger to prevent the charging cable from contacting the floor when not connected to the individual EV. A coupler may be secured to the retracting mechanism and may be configured to easily couple to charging cables of various diameters in a non-slip manner.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Appl. No. 63/559,867.

BACKGROUND

[0002] Electric bikes, scooters, dirt bikes, motorcycles, and other similar small electric vehicles (hereinafter generally referred to as “individual EVs”) have recently become common. These individuals EVs are oftentimes parked in a garage or similar structure where they may be connected to a battery charger. It is typical for the battery chargers to be left hanging from the outlet and onto the floor where they can be stepped on or exposed to dirt and water. It is also common for the battery chargers to be misplaced. These issues are especially prevalent when a child or youth rides an individual EV.

[0003] Additional difficulties exist when a family or location has multiple individual EVs that are all parked near the same outlet. Even if an effort is made to organize the battery chargers, it is typically inevitable that the battery chargers will end up on the floor or misplaced.

BRIEF SUMMARY

[0004] The present disclosure is directed to enclosures for battery chargers for individual EVs. An enclosure can be configured to house a battery charger and can be secured to a wall near an outlet or other source of electricity. The enclosure can include a retracting mechanism such as a badge reel that can be secured to the charging cable of the battery charger to prevent the charging cable from contacting the floor when not plugged into the individual EV. A coupler may be secured to the retracting mechanism and may be configured to easily connect to charging cables of various diameters in a non-slip manner.

[0005] In some embodiments, an enclosure for a battery charger for an individual EV includes a base that is configured to support an adapter of the battery charger and a badge reel that is supported by the base. The badge reel is configured to couple to a charging cable of the battery charger such that the charging cable is retracted when not plugged into the individual EV.

[0006] In some embodiments, the base may form an enclosed area in which the adapter is supported and the enclosure may include a lid that covers the enclosed area.

[0007] In some embodiments, the enclosure may include a coupler that is connected to a retractor of the badge reel and configured to receive the charging cable.

[0008] In some embodiments, the coupler forms a charging cable channel with one or more curves or bends.

[0009] In some embodiments, the charging cable channel is S-shaped.

[0010] In some embodiments, the base includes a retractor opening through which a retractor of the badge reel extends out from the base.

[0011] In some embodiments, the retractor opening is formed through a floor of the base.

[0012] In some embodiments, the floor of the base forms a charging cable opening through which the charging cable extends out from the enclosure.

[0013] In some embodiments, the base includes one or more cord notches.

[0014] In some embodiments, the base includes a cartridge in which the badge reel is housed.

[0015] In some embodiments, the base includes a cartridge housing that receives the cartridge.

[0016] In some embodiments, a retractor opening is formed within the cartridge housing.

[0017] In some embodiments, a charging cable opening is formed adjacent to the cartridge housing.

[0018] In some embodiments, the enclosure may include a coupler secured to a retractor of the badge reel, the coupler including a main housing and a cover and forming a charging cable channel through which the charging cable is routed.

[0019] In some embodiments, the cover is rotatable relative to the main body.

[0020] In some embodiments, the charging cable channel includes two or more curves or bends.
[0021] In some embodiments, a coupler for securing a charging cable of an individual EV to a retractor may include a main body and a cover that form a charging cable channel having one or more curves or bends. The charging cable channel includes an ingress and an egress for the charging cable.

[0022] In some embodiments, the charging cable channel is S-shaped.

[0023] In some embodiments, an enclosure for a battery charger for an individual EV may include a base forming an enclosed area configured to receive an adapter of the battery charger. The base forms a charging cable opening and a retractor opening. The enclosure also includes a retracting mechanism having a retractor that extends out of the enclosure through the retractor opening.

[0024] In some embodiments, the enclosure may include a coupler that is secured to the retractor. The coupler forms a charging cable channel through which a charging cable can be routed when the charging cable extends out of the enclosure through the charging cable opening.

[0025] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] In order to describe the manner in which the above-recited and other advantages and features can be obtained, a more particular description will be rendered by reference to specific embodiments which are illustrated in the appended drawings. These drawings depict only typical embodiments and are not therefore to be considered to be limiting.

[0027] FIGS. 1A and 1B illustrate an example of an enclosure configured in accordance with one or more embodiments of the present disclosure.

[0028] FIGS. 2A-2E illustrate a base of the enclosure of FIGS. 1A and 1B.

[0029] FIGS. 3A and 3B illustrate a lid of the enclosure of FIGS. 1A and 1B.

[0030] FIG. 4 is a cross-sectional view of the enclosure of FIGS. 1A and 1B.

[0031] FIGS. 5A-5D illustrate a coupler that is configured in accordance with one or more embodiments of the present disclosure.

[0032] FIGS. 6A-6D provide an example of how an enclosure configured in accordance with one or more embodiments of the present disclosure may be used.

[0033] FIG. 7 provides an example of how multiple enclosures may be interconnected.

DETAILED DESCRIPTION

[0034] In this specification and the claims, the term “battery charger” will be used to represent the entire charging assembly necessary to plug in an individual EV. A battery charger includes an “adapter” and a “charging cable.” The adapter is the typically rectangular portion of the battery charger that typically converts AC voltage to DC voltage. The charging cable should be construed as the cable that plugs into the individual EV. A battery charger may also include a “cord” which should be construed as the cable that plugs into an outlet. However, some battery chargers have an adapter into which the plug is integrated and therefore such battery chargers do not have a cord.

[0035] FIGS. 1A and 1B provide an example of an enclosure 50 that is configured in accordance with one or more embodiments of the present disclosure. Enclosure 50 includes a base 100 and a lid 200 that couple together to cause enclosure 50 to have a generally rectangular shape. However, an enclosure could be configured in any other suitable shape. Base 100 forms an enclosed area 100a in which the adapter of a battery charger may be placed. Lid 200 can be configured to at least partially cover enclosed area 100a to prevent removal of the battery charger from enclosure 50.

[0036] Enclosure 50 also includes a coupler 300 and a retracting mechanism 400. Retracting

mechanism **400** may be housed within or otherwise supported by base **100** and/or lid **200**. Coupler **300** is connected to retracting mechanism **400** and is configured to couple to a charging cable as described in detail below.

[0037] FIGS. 2A-2E are various views of base **100** in isolation. Base **100** may include walls and a floor that form enclosed area **100a**. Enclosed area **100a** may be shaped and sized to receive an adapter of a battery charger. In some embodiments, a wall of base **100** may include one or more openings **101** (e.g., screw holes) for mounting base **100** to a wall. In some embodiments, base **100** may be configured to be secured to a wall via an adhesive, hook and loop, or another fastening mechanism.

[0038] A cord notch **102** could be formed in the upper edge of one or more of the walls of base **100**. For example, in the depicted embodiments, cord notches **102** are formed in each side wall. Each cord notch **102** can be configured to accommodate the cord of the battery charger, the charging cable of the battery charger, or an extension cord when lid **200** is placed on base **100**.

[0039] In some embodiments, including the depicted embodiments, base **100** includes a cartridge housing **110** and a cartridge **120** that is securable within cartridge housing **110**. Cartridge **120** is configured to house retracting mechanism **400**. For example, cartridge **120** may form an enclosed area **120a** that is shaped and sized to receive the body of a badge reel which may be round, square or another shape. Opposing ribs **122** may extend into enclosed area **120a** and may be configured to be positioned on opposing sides of a clip on the back of the badge reel body to prevent the badge reel body from rotating within cartridge **120**. A bottom of cartridge **120** may form a cartridge opening **121** through which the retractor of the badge reel extends. When cartridge **120** is inserted into cartridge housing **110**, the opening to enclosed area **120a** can face the wall of base **100** to thereby secure the badge reel within cartridge **120**. In other embodiments, however, cartridge **120** may be omitted and retractor **400** may be housed directly in cartridge housing **110**. Accordingly, cartridge housing **110** could have a variety of shapes and/or configurations for receiving/securing retractor **400**.

[0040] A retractor opening **104** may be formed in the floor of base **100** within cartridge housing **110** and, in embodiments that employ cartridge **120**, may align with cartridge opening **121** to allow the retractor of the badge reel to extend from base **100**. A charging cable opening **103** may also be formed in the floor of base **100** to allow the charging cable of the battery charger to extend from base **100** when its adapter is contained in enclosed area **100a**. Charging cable opening **103** may be positioned adjacent cartridge housing **110** so that the charging cable will be positioned adjacent to the retractor of the badge reel. However, in some embodiments, charging cable opening **103** could be positioned anywhere in the floor or a wall of base **100**.

[0041] FIGS. 3A and 3B illustrate lid **200** in isolation. Lid **200** may include a lip **201** that is shaped and sized to insert tightly within base **100**. Lid **200** may also include a cord notch **202** that may be centered along a rear edge. Like cord notches **102**, cord notch **202** can allow the cord, an extension cord, or the charging cable to extend into enclosed area **100a** when lid **200** is in place on base **100**.

[0042] FIG. 4 is a cross-sectional view of enclosure **50** and provides an example of how retracting mechanism **400** can be housed within base **100**. In this example, retracting mechanism **400** is in the form of a badge reel having a body **401**, a clip **402** (not visible), a retractor **403**, and a coupling element **404**. Body **401** is oriented to position clip **402** between ribs **122** and to position retractor **403** downward. In some embodiments, cartridge **120** may have a height that is greater than the vertical dimension of body **401** to allow body **401** to slide within cartridge **120**. This ability of body **401** to slide within cartridge **120** allows body **401** to absorb the force caused when retractor **403** retracts to prevent cartridge **120** from sliding out from cartridge housing **110**.

[0043] FIGS. 5A and 5B are front and rear views respectively of coupler **300**. Coupler **300** includes a main body **310** and a cover **320** that is secured to main body **310**. FIG. 5C is a front view of main body **310** in isolation. FIG. 5D is a rear view of cover **320** in isolation.

[0044] Cover **300** forms a charging cable channel **301** which includes an ingress **301a** and an

egress **301b**. In some embodiments, including the depicted embodiments, main body **310** may be recessed to form this charging cable channel **301**. Charging cable channel **301** may include one or more curves or bends which create friction points for preventing a charging cable routed through charging cable channel **301** from sliding relative to charging cable channel **301** when the charging cable is pulled. In the depicted embodiments, cover **300** is shaped like an S which creates two curves/bends in charging cable channel **301**. However, a cover could include a single curve/bend or more than two curves/bends in some embodiments. In some embodiments, one or more tabs **314** could be formed within charging cable channel **301** to increase the degree of the curve/bend and/or the friction on the charging cable.

[0045] The S-shaped configuration of the depicted embodiments of cover **300** provides the added benefit of preventing the charging cable from bending too much (see e.g., FIG. 6D). In particular, the S-shape with the widened middle section between the curves ensures that the bend/curve radius of the charging cable is maximized to prevent damage to the conductors while still providing sufficient friction to prevent the charging cable from sliding within charging cable channel **301**.

[0046] In some embodiments, main body **310** and cover **320** can be configured to enable cover **320** to be rotated relative to main body **310**. For example, in the depicted embodiments, main body **310** forms a receiver **312** and cover includes a post **321** that inserts into and rotates within receiver **312**. In some embodiments, post **321** may include tabs **321a** and receiver **312** may include corresponding notches **312a** to accommodate tabs **321a**. Accordingly, tabs **321a** and notches **312a** can be aligned to allow post **321** to be inserted into receiver **312**. This alignment can occur when cover **320** does not align with main body **310**. In some embodiments, the bottom of main body **310** may include a recess **313** into which tabs **321a** may extend and may rotate. As shown in FIG. 5B, this rotation causes tabs **321a** to interlock with main body **310** to prevent cover **320** from being separated from main body **310**, particularly when cover **320** is aligned with main body (i.e., while cover is in use).

[0047] In some embodiments, cover **320** can include ribs **322** that insert into cutouts **314** in main body **310** to prevent the unintentional rotation of cover **320** relative to main body **310**. In some embodiments, ribs **322** and cutouts **314** may be positioned towards ingress **301a** and egress **301b** to minimize flexing of cover **320** relative to main body **310**. In such embodiments, cover **320** can be rotated relative to main body (e.g., to allow the charging cable to be removed) by first lifting the ends of cover **320** to remove ribs **322** from cutouts **314** (see, e.g., FIG. 6D).

[0048] Main body **310**, or possibly cover **320**, may form a coupling structure **311** by which cover **300** is secured to retracting mechanism **400**. For example, coupling structure **311** could be a loop structure (e.g., as shown), a cutout, a lip, a magnet, etc.

[0049] FIGS. 6A-6D provide an example of how enclosure **50** may be used. In this example, enclosure **50** is mounted to a wall below an outlet **510**. An adapter **531** of a battery charger for a scooter **520** is housed within enclosure **50** with its cord **532** routed out through cord notch **202** and plugged into outlet **510** and its charging cable **533** extending out through charging cable opening **103**. Charging cable **533** has been routed through coupler **300** which is connected to retracting mechanism **400**. In FIG. 6A, charging cable **533** is plugged into scooter **520** such that retractor **403** is extended. In FIG. 6B, charging cable **533** has been unplugged and retractor **403** has been retracted. Because coupler **300** is fixed to charging cable **533**, charging cable **533** is automatically retracted and held off the floor.

[0050] As can be seen, enclosure **50** organizes the battery charger and simplifies the process of plugging in and unplugging the charging cable. Enclosure **50** can also incentivize riders to park their individual EVs in the desired location. For example, if a father wants his children to park their scooters along the garage wall, enclosure **50** can require them to do so if the scooters need to be plugged in. Enclosure **50** can therefore greatly enhance a parent's efforts to keep a garage organized. Furthermore, in some embodiments, enclosure **50** may be sized to accommodate more than an adapter within enclosed area **100a**. For example, many individual EVs require frequent

maintenance using small and/or specialized tools (e.g., a hex wrench, a tire stem extension, etc.). Such tools may be stored in enclosure **50** where they will be easily accessible when needed. [0051] In some embodiments, base **100** may include one or more hooks or other attachment structures to allow an individual to store items by hanging them from base **100**. For example, a hook could be formed on the side of the base opposite cartridge **120** to allow a lock, bag, or other item to be stored when not in use.

[0052] In some embodiments, retracting mechanism **400** may be a retracting spool rather than a badge reel. In such cases, the charging cable can be wrapped around the retracting spool such that only its end extends out through charging cable opening **103** when retracted. In other words, in embodiments in which a retracting spool is provided, the charging cable may be fully retracted into enclosure **50** as opposed to hanging beneath enclosure **50**.

[0053] In some embodiments, cord notches **102** can facilitate daisy chain connections between adjacent enclosures **50**. For example, as represented in FIG. 7, a family having multiple electric scooters may mount multiple enclosures **50** on a garage wall where they desire that the scooters be parked for charging. Then, to avoid having to connect all the battery chargers separately to an outlet or power strip, a first extension cord **601** could be routed into one of enclosures **50** (e.g., via cord notch **202** or via one of cord notches **102**). This first extension cord **601** could include two sockets into which the cord of the battery charger in the enclosure **50** may be plugged in. A second extension cord **602** could be plugged into the other socket and routed out through cord notch **102** and then in through cord notch **102** of an adjacent enclosure **50**. The second extension cord **602** could also include two sockets to allow the cord of the battery charger in that enclosure and another extension cord **603** (or the cord of another battery charger) to be plugged in. This type of daisy chain connection between adjacent enclosures **50** could be repeated as necessary to provide power inside all enclosures **50** via the first extension cord **601**.

Claims

1. An enclosure for a battery charger for an individual electric vehicle (EV) comprising: a base that is configured to support an adapter of the battery charger; and a badge reel supported by the base, the badge reel being configured to couple to a charging cable of the battery charger such that the charging cable is retracted when not plugged into the individual EV.
2. The enclosure of claim 1, wherein the base forms an enclosed area in which the adapter is supported, the enclosure further comprising: a lid that covers the enclosed area.
3. The enclosure of claim 1, further comprising: a coupler that is connected to a retractor of the badge reel and configured to receive the charging cable.
4. The enclosure of claim 3, wherein the coupler forms a charging cable channel with one or more curves or bends.
5. The enclosure of claim 3, wherein the charging cable channel is S-shaped.
6. The enclosure of claim 1, wherein the base includes a retractor opening through which a retractor of the badge reel extends out from the base.
7. The enclosure of claim 6, wherein the retractor opening is formed through a floor of the base.
8. The enclosure of claim 7, wherein the floor of the base forms a charging cable opening through which the charging cable extends out from the enclosure.
9. The enclosure of claim 1, wherein the base includes one or more cord notches.
10. The enclosure of claim 1, wherein the base includes a cartridge in which the badge reel is housed.
11. The enclosure of claim 10, wherein the base includes a cartridge housing that receives the cartridge.
12. The enclosure of claim 11, wherein a retractor opening is formed within the cartridge housing.
13. The enclosure of claim 12, wherein a charging cable opening is formed adjacent to the cartridge

housing.

14. The enclosure of claim 1, further comprising: a coupler secured to a retractor of the badge reel, the coupler including a main housing and a cover and forming a charging cable channel through which the charging cable is routed.

15. The enclosure of claim 14, wherein the cover is rotatable relative to the main body.

16. The enclosure of claim 14, wherein the charging cable channel includes two or more curves or bends.

17. A coupler for securing a charging cable of an individual EV to a retractor, the coupler comprising: a main body and a cover that form a charging cable channel having one or more curves or bends, the charging cable channel including an ingress and an egress for the charging cable.

18. The coupler of claim 17, wherein the charging cable channel is S-shaped.

19. An enclosure for a battery charger for an individual EV comprising: a base forming an enclosed area configured to receive an adapter of the battery charger, the base forming a charging cable opening and a retractor opening; and a retracting mechanism having a retractor that extends out of the enclosure through the retractor opening.

20. The enclosure of claim 19, further comprising: a coupler that is secured to the retractor, the coupler forming a charging cable channel through which a charging cable can be routed when the charging cable extends out of the enclosure through the charging cable opening.
